Pattern Mining of Older Drivers' Driving Behavior Through Telematics-Data-Driven Unsupervised Learning

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***Abstract*—Increasing age and age-related health conditions can contribute to a higher risk of crash involvement and more severe physical injuries among older drivers. The current study uses in-vehicle sensing systems to analyze and cluster the overall driving behaviors of older drivers, precisely those aged 65 and older. The primary objective is to develop a framework that can help to observe patterns indicative of distinctive driving styles by clustering and interpreting similar patterns and leveraging self-organizing maps (SOMs) and deep embedded clustering (DEC) methods to reduce the complexity of the sensor data. The research uses two distinct groups of variables, including "speed," "hard acceleration," "hard braking" as constant variables, and "rpm," "throttle positioning," "fuel level," "engine level," and "ambient air temperature" as target variables from the in-vehicle sensor data in understanding driving behaviors and develops models in visualizing and interpreting complex driving patterns by classifying them. The results show that 5 × 5 grid SOMs have the ability to visualize multiple driving features concurrently, and DEC + K-means and DEC + agglomerative are the well-performed methods for determining the optimal number of clusters to analyze types of driving patterns. The clustering analysis identifies two distinct clusters as the optimal configuration, indicating that the predominant driving behavior among the targeted participants exhibits conservative patterns. The methodologies can be used for other driving features and demographics, making it relevant for broader applications in traffic analysis for understanding and visualizing driving patterns.**

***Index Terms*—Deep embedded clustering (DCE), driving behavior, older drivers, self-organizing maps (SOMs), sensor data.**

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I. Introduction

PEOPLE aged 65 and older comprised 17.3% of the population in 2022. This proportion is projected to increase to 22% by the year 2040 [1]. According to the National Highway Traffic Safety Administration (NHTSA), drivers aged 65 and older were involved in 7870 fatal accidents in 2022, constituting 20% of all fatal traffic collisions [2]. Many older adults value the independence of driving, but age-related changes and common health conditions, including medication side effects, can impair their ability to drive safely [3]. With advancing age, cognitive decline becomes prevalent, and dementia may represent an extreme on this continuum [4]. This decline is associated with reduced driving ability and less driving exposure among older adults [5]. Older drivers face increased physical vulnerability and declines in sensory, perceptual motor, and cognitive abilities, which can be compensated for but may break down in complex traffic situations [6]. Declines in physical and perceptual functioning compromise the driving performance and safe mobility of older adults [7]. Driving cessation in older adults is associated with increased depression and social isolation, highlighting the need to enhance their safety while preserving mobility [8].

The Federal Highway Administration (FHWA) reports demonstrate that driving behaviors, such as speeding, aggressive lane changes, and inattention to the road, contribute to a significant proportion of road incidents [9]. The World Health Organization (WHO) has consistently reported on the global impact of road traffic accidents, declaring them a leading cause of death and a major public health concern [10]. Therefore, the necessity of data-driven approaches for behavioral analysis and evaluation of the effective factors related to drivers is important to mitigate risky driving behaviors and enhance road safety [11], [12]. Analyzing drivers' actions and decisions that affect road safety and accurate assessment methods help pinpoint real-time risks for proactive interventions and preventive measures, and implement them quickly and successfully [13]. Early studies focused on fundamental statistical models to assess the risk associated with driving behaviors, relying on subjective evaluations of human observers and hands-on analysis of recorded data [14], [15].

Recent studies have concentrated on the capability of telematics, which integrates different technologies to offer comprehensive data on drivers' behavior [16], [17]. The term "telematics" has denoted the fusion of informatics and telecommunications, which has recently been referred to as "automotive telematics," using telecommunications and informatics to improve vehicles' capability through the global positioning system (GPS), intelligent cruise control, and wireless data applications [18], [19]. Rapid advancement in telematics has transformed the landscape of vehicle data analytics, offering profound insight into driving behaviors [20]. Some studies find telematics to be a safe and comfortable method for transportation systems with access to vast information on trips [21]. Telematics sensors capture a wealth of information that can be analyzed to identify patterns and predict risk factors associated with driving behaviors [22]. Telematics sensor data, which encompasses a broad spectrum of metrics such as speed, acceleration, and braking patterns, provide fertile ground for the application of advanced data analysis techniques [23].

On the other hand, the evolution of artificial intelligence and machine learning algorithms has provided new pathways for telematics data analysis [24], [25], [26]. Machine learning techniques enable predictive models to participate in driving patterns with historical data to detect risky and aggressive driving behaviors related to speed, acceleration, lane changes, and near-collision events [27], [28]. Recent studies demonstrate the importance of indicators such as acceleration based on trajectory data to identify abnormal driving patterns [29]. Using accelerometer-related data improves the identification of driving behaviors through machine learning techniques [30]. Clustering and pattern recognition algorithms applied to an extensive dataset of driving behaviors to establish distinct behavior clusters, such as aggressive, cautious, and everyday driving, with clear correlations established between them and accident risk levels [31]. Different neural network approaches were used to understand driving behavior [32]. Some neural network models show a high performance in predicting features like acceleration distribution [33]. However, for high-dimensional datasets, these methods are inadequate for visualizing the features associated with driving behavior to gain a detailed understanding.

Over the past few decades, several foundational studies in naturalistic driving have used statistical modeling techniques to gain insight into driving behaviors. However, these early statistical approaches encountered limitations that restricted their effectiveness in comprehensively analyzing driving behaviors. One significant challenge is the inherent restriction of linear mixed models, which may not accurately represent the complex, nonlinear interactions in driving data. Additionally, issues such as over-dispersion in the data—where the variance exceeds the mean—often led to inefficient and biased estimates, undermining the reliability of the findings. Further limitations included incomplete solutions in scenarios with sparse data, a common issue in earlier driving studies where long-term monitoring is not feasible. Moreover, the scalability of these models is a critical concern; as the volume and dimensionality of telematics data grew with advances in sensor technology, the traditional statistical models struggled to process large-scale datasets efficiently [34], [35], [36], [37]. Most previous research has depended on data collected from smartphones, surveys, field experiments, and simulations. However, how people drive in actual situations frequently varies from the behaviors observed in simulated environments. Additionally, using smartphone data to analyze driving behavior has some challenges, such as low accuracy, the positioning of the smartphone, and data inconsistency due to the driver's interaction with the device and battery usage. Also, smartphones are limited to data from GPS and built-in accelerometers [38], [39].

Our study develops an in-vehicle sensing platform aimed at identifying driving style patterns among older drivers using cutting-edge machine learning techniques. The research addresses several challenges, including integrating and correlating data from diverse in-vehicle sensors like GPS, inertial measurement unit (IMU), and on-board diagnostic (OBD) systems. These sensors generate substantial data, posing significant hurdles in data processing and analysis to determine driving behavior accurately. Our methodology uniquely applies self-organizing maps (SOMs) and deep embedded clustering (DEC) to overcome these issues. SOMs help visualize and correlate features, enhancing understanding of driving behavior. At the same time, DEC, used with clustering techniques like K-means and Gaussian mixture model (GMM), evaluates the coherence of these behaviors in older drivers. Similarly, our approach determines the optimal number of clusters using U-matrix and various validity indices, ensuring precise categorization of driving patterns. This framework not only contributes to academic research by offering a nuanced analysis of older drivers' behaviors but also enhances real-time monitoring solutions, ensuring their safety and accommodating their specific needs. The study's findings are tailored to the unique characteristics and needs of drivers aged 65 and older, derived from extensive data collected over three years, highlighting its specificity and relevance to this demographic.

A. Literature Review

Among unsupervised learning techniques, SOM, a type of machine learning method, has gained prominence due to its ability to map high-dimensional data into lower dimensional spaces while preserving data topology [40]. For the first time, Kohonen [41] introduced SOM, an unsupervised artificial neural network, as a machine-learning technique to learn and categorize complex patterns in high-dimensional data. SOM helps to develop systems that monitor and map human activities [42], [43]. SOM showed an efficient approach for visualizing and confirming correlations between different materials, displaying both numerical attributes and categorical information associated with materials [44]. Many studies in the fields of medicine, geoscience, and imaging have found that SOM is a reliable tool to visualize, use multiple functional class labels, and cluster connectivity maps associated with different features [45], [46], [47], [48], [49], [50]. SOM has proven effective in identifying driving patterns using telematics data gathered through smartphones [38]. Comparative analysis using SOM and K-means clustering showed a good performance in identifying distinct multivariate behavior patterns across driver groups, aiding policy decisions and resource allocation for improved road safety [51]. Moreover, the integration of the SOM-K-means algorithm can efficiently enable the clustering of drivers showing distinct driving styles [52]. SOM produced a highly interpretable solution with high accuracy in identifying and classifying driving maneuvers [53]. Using SOM for clustering OBD data could help to identify the driving style and degree of danger [54]. The integration of autoencoder and SOM (AESOM) was applied to GPS data to evaluate the effectiveness of SOM for analyzing and drawing insight about driving behavior like speed and acceleration [55].

With all the advantages of SOM, several limitations have been reported in using this method. First, SOMs face challenges in vector quantization and multidimensional scaling compared to techniques combining K-means clustering with Sammon mapping. This comparison highlights that while SOMs are effective for visual pattern recognition and data dimensionality reduction, they may not perform well in preserving the precise distances between high-dimensional data points as well as the combined approach. Furthermore, the traditional static architecture of SOM limits its ability to represent hierarchical data relations effectively. Another significant limitation of SOM concerns the computational time cost, which can be substantial, particularly with large datasets. Inconsistent or inadequate normalization can lead to suboptimal clustering results, as the quality of SOM learning and output is highly dependent on the initial presentation of data. Normalization mechanism enhancement could mitigate this limitation, improving both the robustness and performance of SOMs in handling diverse and complex datasets [56], [57], [58].

Meanwhile, clustering remains an important technique in machine learning and data mining, gaining significance mainly through the advent of deep learning methodologies. DEC is a prominent approach among deep clustering techniques, offering advanced solutions for diverse clustering tasks [59]. The DEC method was first introduced by Xie et al. [60]. They present a methodology that simultaneously learns feature models and clustering using deep neural networks. DEC optimizes clustering objectives by learning a mapping from the data space to a lower dimensional feature space, where it iteratively refines both the cluster centroids and the feature representation of the data [60]. Recently, using different machine learning and deep learning approaches to extract distinguished patterns in driving data and comparing K-means clustering, hierarchical clustering, and GMM to study driving behavior characteristics has become a common approach in this field [61], [62]. Some studies propose the hierarchical clustering method for extracting driving performance [63]. On the other hand, most of the studies find K-means clustering as the strongest method in classifying driving data [64], [65], [66]. A deep learning approach could have successfully detected different types of driving behavior and improved the detection performance [67], [68].

One key limitation of DEC is its inability to utilize prior knowledge to guide the learning process. Existing DEC methods often struggle due to the lack of labeled data, which is critical for learning discriminative features in supervised learning environments. This limitation can hinder the model's ability to effectively distinguish between nuanced driving behaviors, particularly in datasets with subtle and complex behavior patterns. Additionally, spectral clustering, while valuable for identifying community structures within data, faces challenges regarding scalability. The computational complexity of spectral clustering increases significantly with the size of the dataset, making it less feasible for large-scale applications, such as those involving extensive telematics data. This lack of scalability can restrict its use in real-time or near-real-time applications, where quick data processing is essential. Another notable issue with spectral clustering is the lack of generalization in the spectral embedding, particularly concerning out-of-sample extensions, which means that the method often fails to perform well when applied to new, unseen data, limiting its utility for predictive modeling where the ability to generalize to new instances is crucial. These limitations necessitate careful consideration when deploying these methods in practical settings and highlight the need for ongoing improvements in clustering technologies to better handle the complexities of telematics data [59], [69], [70].

B. Problem Definition

In this research, the first step aims to map driving behavior to observe and predict whether the driving behavior aligns with distinctive driving based on data obtained from the programmable in-vehicle telematics unit. We select the features that potentially indicate aggressive driving behavior based on previous studies in this field, including speed and acceleration [71]. Mapping these features helps to see the frequency with which drivers often cause higher values of these features in their everyday driving activities, which might tend to aggressive driving behavior. The next phase involves comprehensively analyzing the data using different methods to classify the dataset according to its values. Previous research has utilized simpler clustering methods to analyze naturalistic driving data and classify driving behaviors [72]. Our study extends this approach by employing SOM, which allows for simultaneous visualization of multiple driving features and their interactions and values in the form of a spectrum. This capability provides an intuitive and holistic view of driving behavior patterns. Additionally, we incorporate DEC, which combines simpler clustering techniques to validate and enhance cluster determination, ensuring robust classification results from SOM. In summary, we present two definitions for these steps.

1) Definition1- Holistic Analysis: This approach considers the entirety of SOMs, including all their features and interactions, to understand the emergent properties and behaviors arising from these interactions. With this approach, we focus on identifying the features with the highest value at this stage, and analyze the combination of concurrent features adjacent to each other on the map. In certain instances, a decrease in the quantity of a feature and an increase in the quantity of another feature may serve as indicators of distinct behavior. In other words, this approach provides us with an overview of the driving behavior of this age group without the complexity of numbers or detailed examination, allowing for rapid and clear comprehension.

2) Definition2- Component-Level Analysis: This concept involves examining each component (here as a feature) solely to understand its properties, behaviors, and potential interactions with other features. Two distinct approaches are taken for this concept. Initially, we used the U-matrix for each SOM to display the clusters, determine the optimal number of clusters, and describe the characteristics of each cluster. The second strategy involves comparing the clustering results of the first approach with DEC. This comparison serves to determine the characteristics of the clustered data and to create the most accurate assessment of driving behavior analysis. We used three different clustering methods: K-means, agglomerative clustering, and GMM for DEC. We determined the number of clusters based on the most effective performance metrics and consolidated the cluster evaluation for the most optimal method.

II. Method and Design

A. Programmable Telematics Units

This study leverages a TMU developed by AutoPi [73], which builds on Raspberry Pi 4 Model B and provides an open-source platform. These programmable and open-sourced TMUs support the robust system expandability of hardware and software, allowing the development of customized TMUs to adjust and modify the sensing parameters (e.g., data collection algorithms and sampling frequencies). Each TMU comprises: 1) a GPS sensor; 2) an IMU sensor; 3) an OBD connector; 4) a 4G/LTE cellular modem; 5) an SD card; and 6) a USB flash drive (see Fig. 1).

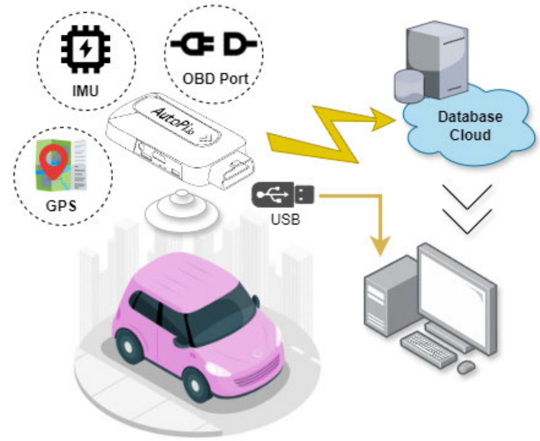

Fig. 1. Programmable telematics units overview.

Tri-axial gyroscopes are added to TMUs to capture angular velocity in degrees per second. The TMU has a smart power system. It continuously monitors the voltage levels from the port. When the voltage goes down to 12V, it automatically goes into sleep mode for five minutes. After waiting for another five minutes, the device goes into hibernation mode. For the AutoPi telematics unit, the voltage typically drops to 12V under conditions such as when the vehicle is stopped or parked. This voltage drop occurs because the engine is off. Thus, the alternator does not generate additional voltage, causing the system to rely solely on the battery's standing voltage. A fully charged automotive battery will have a standing voltage of about 12.6V, but this can drop to around 12V or slightly less under load when lights or other electronics are used while the vehicle is off. The duration of the 12V condition depends on several factors, including the battery's health, the external temperature, and any electrical devices in use. If the vehicle remains parked without starting for extended periods, the voltage may decrease as the battery discharges. However, during normal conditions where the vehicle is stopped temporarily (e.g., at traffic lights), the voltage might briefly dip to around 12V and then return to normal (approximately 13.5–14.5V) once the engine is restarted. The alternator begins charging the battery again [73].

The power usage of the AutoPi in its sleep and hibernation mode is as follows: around 30 and 10 mA. In-vehicle data can be collected from the controller area network [74], [75] and augmented sensor hardware. One of the most common ways of obtaining CAN bus data is using OBD connectors. Available OBD data include engine RPM, vehicle speed, and fuel system status. In addition to the OBD data, additional data can be collected from augmented sensor hardware, such as GPS and IMU modules (see Fig. 2). Furthermore, various unobtrusive sensing units can be installed on a vehicle to understand the state of drivers [76]. Fig. 3 shows an actual sensor installation on a participant's vehicle, providing real-world telematics data for this study.

B. Data Collection

This research project is designed to gather ongoing data regarding the driving patterns of older drivers, which is

Fig. 2. IMU overview.

Fig. 3. In-vehicle sensor deployment.

TABLE I
DEMOGRAPHIC INFORMATION OF PARTICIPANTS

Demographic Info	Categories	Numbers
Language	English	226
	Spanish	18
	Creole	24
Gender	Female	159
	Male	109
Retired	Yes	207
	No	61
Marital Status	Married	128
	Divorced	44
	Widowed	60
	Single	36
Race	Black	39
	White	217
	Asian	3
	Native American	0
	Multiracial	4
	Other	5
Ethnicity	African American	18
	European American	187
	Hispanic American	18
	Afro-Caribbean	22
	Asian American	2
	Other	20
Bilingual	Yes	45
	No	223
Handedness	Right	235
	Left	24
	Ambidextrous	9
Highest Degree	Grade School	7
	High School	23
	Some College	32
	Technical School	3
	Associate's Degree	19
	Bachelor's Degree	87
	Post-Graduate Studies	15
	Master's Degree	53
	Post-Master's Studies	7
	Doctoral Degree	22

Demographic Info	Min	Mean	Max
Age	65	74.88	91
Height (inch)	50	65.98	78
Weight (lb)	86	166.38	280
BMI*	15.72	27.22	48.05
Years of Education	0	16.11	26

*BMI = Weight/(Height)^2

analyzed alongside thorough evaluations conducted quarterly over three years to determine the specific driving safety requirements for older drivers across the United States. Participants must be aged 65 or above, hold a valid driving license, have car insurance, and operate either a sedan or a pickup truck. Participants must attend follow-up sessions every three months at one designated location in Boca Raton, Davie, or the University of Central Florida (UCF) to facilitate data collection, sensor checks, and necessary maintenance. Table I briefly explains the participant's demographic information.

Eligibility further demands that candidates be proficient in English, Spanish, or Haitian Creole and meet Florida's standard requirements for drivers regarding vision, hearing, strength, and flexibility. The study recruitment process is spearheaded by an experienced recruiter who establishes connections with community organizations serving seniors, including residential facilities for older individuals, religious institutions, and medical centers. Outreach initiatives are supported by bilingual staff to accommodate nonEnglish speakers, and informational materials are provided in the three aforementioned languages. An approved Institutional Review Board participant referral program also offers gift card rewards for successful referrals. Prospective participants undergo a screening process, and upon meeting all requirements and consenting to participate, initial assessments are carried out. This is followed by the installation of driving behavior monitoring sensors in their vehicles. At the time this report was compiled, 268 participants had been enrolled in the study. About 51 participants were selected as they entered the study under uniform conditions, which mitigates potential biases in

TABLE II
FEATURES FROM TELEMATICS UNIT AND STATISTICAL OVERVIEW

Index	Description	Max	Mean	Standard Deviation (STD)
Hard Accelerations	Unit: No. Number of events with acceleration $> 0.3g^*$ on the X-axis	24	0.88	1.32
Hard Brakings	Unit: No. Number of events with acceleration $< -0.3g^*$ on the X-axis	44	1.20	1.60
Speed	Unit: km/h Speed in kilometers per hour	144.33	45.47	22.59
RPM	Revolutions per minute	16383.75	1622.48	548.10
Engine Load	Unit: % Percentage of engine load	100	54.29	16.37
Ambient Air Temperature	Unit: °C Air temperature in the environment	70.29	30.18	9.22
Fuel Level	Unit: % Percentage of fuel level	100	71.74	22.52
Throttle Positioning	Unit: % Percentage of throttle positioning	100	26.01	12.08

$^*g$: the acceleration of gravity (9.81 m/s^2)

the analysis and ensures a more controlled evaluation of the data.

Telematics data include multitype data from accelerometers, gyroscopes, GPS navigation systems, and OBD communications. These data include the location (latitude, longitude, course over ground, and altimeter); speed; three-axis acceleration; events related to acceleration, such as hard braking and hard turns; rpm (revolutions per minute); and trip start/end (number of trips, time, and distance). This study used the 19 881 number of trips with a cumulative distance of 23612941.55 km collected periodically or in response to specific aggregation over three years for 51 drivers aged 65 and above with more than 30 features, and each data point represents one trip. According to Table II, seven numerical variables are extracted from telematics data for this study to reflect certain aspects of driving behavior, including "speed," "hard acceleration," "hard braking" as constant variables, and "rpm," "throttle positioning," "fuel level," and "engine level" as target variables and inputs denoted by the matrix of $X \in \mathbb{R}^{n\times p}$ as 1, which are placed separately and four by four in each of the models (n is the number of data points and p indicates the number of features)

$$\boldsymbol{X} = \begin{bmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{\text{np}} \end{bmatrix}. \tag{1}$$

The explanation for selecting "speed," "hard acceleration," and "hard braking" as constant variables is related to their key role in determining aggressive driving behavior [77], [78]. The selection of other telematics-based features (RPM, throttle positioning, fuel level, engine load, and ambient air temperature) is based on prior literature and empirical evidence highlighting their importance in characterizing driving behaviors. We aim to analyze the correlation of these features in comprehending the driving patterns [79], [80]. We have adopted a 0.3 g threshold to classify "hard acceleration" events based on established automotive research that recognizes accelerations exceeding this value as significant instances of aggressive driving. This threshold is widely used in vehicular telematics to ensure consistency and comparability of results across studies. The decision to use a 0.3 g threshold is grounded in empirical evidence suggesting that such acceleration levels are not typical for normal driving behavior and indicate increased risk behaviors. Traffic safety and driver behavior analysis studies have shown that accelerations above this level are often associated with higher accident rates and can be critical indicators of risky driving practices [81].

All experiments are conducted on a workstation with an Intel[1] Xeon[1] W-2225 CPU (four cores, eight threads, 4.10 GHz), 32 GB RAM, and an NVIDIA T1000 GPU with 8 GB VRAM. The implementation is carried out in Python 3.11.7. The system ran on Windows 10. Before data analysis, specific steps are utilized to guarantee data quality and consistency. In the preprocessing phase, we refine our sensor data to enhance data quality and consistency. First, we tackle missing values using median imputation, which replaces gaps with the column's median, mitigating the impact of outliers common in telematics data. Next, we cap extreme outliers in each column at 1.5× the interquartile range (IQR) to minimize skew from sensor errors or unusual driving behaviors. Additionally, we clean the dataset by trimming whitespace and converting data types to numeric, using coercion for unconvertible values. Finally, we apply a standard scaler to all datasets, standardizing features by removing the mean and scaling to unit variance. This ensures uniform contribution of input features to the analysis and prevents bias toward variables with larger scales.

[1]Trademarked.

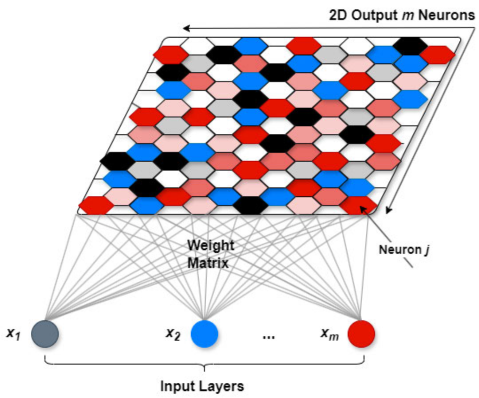

Fig. 4. Overview of a SOM.

C. Self-Organizing Map

SOM, introduced by Kohonen, is an unsupervised machine-learning algorithm that projects high-dimensional data onto a lower dimensional grid while preserving the topological properties of the original space. The mathematical fundamentals of SOMs build upon competitive learning and neighborhood functions. SOM of each model is denoted by m neurons, each associated with a weight vector w_i for ith neuron in n-dimensional input space, where $i = 1, 2, \ldots, m$ (see Fig. 4). m is the total number of neurons allocated for each model ($m \in n$). Initialization of the weight vectors is done randomly or based on some predefined rules, considering (2) as follows:

$$W_{x_j} = \{w_i \in \mathbb{R}^m : i = 1, 2, \ldots, m\}. \tag{2}$$

For the matrix of X, each column is considered as an input vector and for j^{th} neuron ($j = 1, 2, .., m$) represented as the following equation:

$$\overrightarrow{\mathrm{x_j}} = \left[x_1, x_2, \ldots, x_j\right]^{\mathrm{T}}. \tag{3}$$

The superscript T indicates the transpose operation, which is used because vectors are typically expressed as column vectors. For each input vector x_j from the input dataset X, every neuron in the SOM computes its similarity with w_i. This similarity is often calculated using the Euclidean distance as the following equation:

$$\mathrm{d}\left(w_i, x_j\right) = \left\|w_i - x_j\right\|^2 \tag{4}$$

where $\|\|$ denotes the Euclidean norm. The neuron with a weight vector closest to x_j (smallest Euclidean distance) is known as the best matching unit (BMU) and mathematically can be defined as the following equation:

$$i^* = \arg_{\min_i} \mathrm{d}\left(w_i, x_j\right) \tag{5}$$

where $\arg_{\min_i}$ denotes the argument that minimizes the function. i^* is the index of the neuron whose weight vector w_i is the closest to the input vector x_j, making it the BMU for that input. The weights of the BMU and its neighboring neurons are adjusted to become the closest to the input vector x_j. The weight adjustment can be defined as the following equation:

$$w_{i_{(t+1)}} = w_{i_{(t)}} + a_{(t)} \times \mathrm{h}\left(i, {i^*}_{(t)}, s_{(t)}\right) \times \left(x_{j_{(t)}} - w_{i_{(t)}}\right) \tag{6}$$

where $w_{i_{(t+1)}}$ denotes the weight of the ith neuron, $a_{(t)}$ defines the learning rate, and $x_{j_{(t)}}$ is the input vector at time t. On the other hand, $\mathrm{h}(i, {i^*}_{(t)}, s_{(t)})$ is the neighborhood function to determine the influence of the update to the BMU on the ith neuron. The neighborhood function is typically a Gaussian function of the distance between the ith neuron and the BMU, following equation:

$$\mathrm{h}\left(i, {i^*}_{(t)}, s_{(t)}\right) = \exp\left(\frac{-\left\|w_i - x_j\right\|^2}{2 \times s_{(t)}^2}\right) \tag{7}$$

where r_i and r_{i^*} are the positions of the i^{th} neuron and the BMU in the grid, respectively $s_{(t)}$ defines the width of the neighborhood at time t. The learning rate $a_{(t)}$ and neighborhood width $s_{(t)}$ are usually initialized to a significant value and gradually decay over time, which can be represented mathematically as the following equations:

$$a_{(t)} = a_0 \times \exp\left(\frac{-t}{T_1}\right) \tag{8}$$

$$s_{(t)} = s_0 \times \exp\left(\frac{-t}{T_2}\right) \tag{9}$$

where a_0 and s_0 denote the initial learning rate and neighborhood width, respectively, T_1 and T_2 are time constants. The process of competition and adjustment is iteratively performed for several epochs or until the weight change falls below a predefined threshold. The chosen threshold is validated statistically by monitoring the quantization error (QE) and topographic error (TE) as training progresses. The threshold at which these errors plateau indicates an optimal point where additional training does not yield significant improvement in error metrics. In conclusion, the SOM can learn and map high-dimensional driving behavior data categorized into different driving patterns. Each neuron in the trained SOM represents a cluster of similar driving behaviors.

1) Performance of SOM Model:

a) Quantization Error: This error measures how accurately a map represents its training data. This value represents the average distance between each data point and its nearest neuron (BMU). A lower QE means a superior representation of data by SOM; however, a large value of the error indicates an inaccurate representation by the map. The QE in the SOM is computed as the average distance between each data point and its corresponding BMU in the map as the following equation:

$$\mathrm{QE} = \frac{\sum_{i=1}^{p} \mathrm{d}\left(w_i, x_j\right)}{p}. \tag{10}$$

b) Topographic Error: This error refers to the proportion of data points with no adjacent BMUs and ranges between 0 and 1. Lower values generally indicate better SOM performance at maintaining topology within input data. The TE is defined as the following equation:

$$\mathrm{TE} = \frac{\sum_{i=1}^{p} t\left(x_j\right)}{p} \tag{11}$$

where the function $t(x_j)$ is as the following equation:

$$t\left(x_j\right) = \begin{cases} 1, & \text{if } {i^*}_{x_k} \text{ and } {i^*}_{x_{(k+1)}} \text{ are not adjacent} \\ 0, & \text{otherwise.} \end{cases} \tag{12}$$

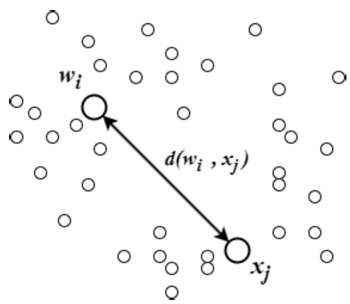

Fig. 5. Average distance between the node and adjacent node.

2) U-Matrix: The unified distance matrix (U-matrix) is an effective visualization approach to interpreting SOMs and potential clusters in each model. Each cell in the U-matrix represents a node (or neuron) of the SOM. The value of the cell is determined by averaging the distances between the node and its adjacent nodes in the input space of high dimension (see Fig. 5). These distances are Euclidean distances $d(w_i, x_j)$ calculated in the input feature space, as discussed in equation (4). High values in the U-matrix indicate a larger distance or dissimilarity between neurons, suggesting that they belong to different clusters or classes. Low values suggest the neurons are similar and likely in the same cluster. The U-matrix is visualized using color maps, where the color scale represents distances and contour lines to show the topology of the map and make the cluster boundaries more explicit [82], [83].

D. Deep Embedded Clustering

DEC is an advanced clustering method that combines autoencoders for dimensionality reduction with clustering. The primary goal of DEC is to learn feature representations that are advantageous for clustering [60]. Following is a breakdown of the algorithmic framework used in DECa.

An autoencoder is trained to learn compressed (encoded) representations of the data. The autoencoder consists of two parts (see Fig. 6).

1) *Encoder:* Maps the input data X to a lower-dimensional representation z as the following equation:

$$z = f_\theta(X) \tag{13}$$

where f_θ denotes the encoder function parameterized by θ.

1) *Decoder:* Attempts to reconstruct the input data from the compressed form z as the following equation:

$$X' = g_\phi(z) \tag{14}$$

where g_ϕ denotes the decoder function parameterized by ϕ.

The autoencoder is trained to minimize the reconstruction loss, typically the mean-squared error (mse) as the following equation:

$$L = \frac{1}{n}\sum_{i=1}^{n} \left\| x_i - g_\phi\left(f_\theta\left(x_i\right)\right)\right\|^2. \tag{15}$$

After training the autoencoder, the encoder transforms all data points into the compressed space, which is named the latent space. The dimensions of the latent space are considerably

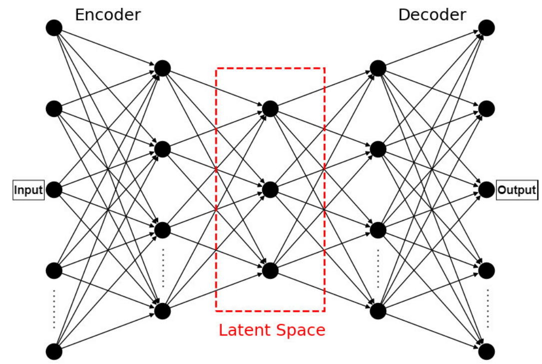

Fig. 6. Deep auto-encoder.

fewer than the input data, but are engineered to retain the characteristics necessary for effective clustering. The initial cluster centers are then calculated using clustering algorithms on these embedded points. This study used K-means, agglomerative clustering, and GMM as clustering algorithms.

1) K-Means: K-means clustering is a method for partitioning data into K distinct groups, optimizing the placement of centroids to minimize within-cluster variance. The process begins by randomly selecting K centroids from the dataset. Each data point is assigned to the nearest centroid, thus forming K initial clusters. The centroids are recalculated by averaging the data points in each cluster, and this step is repeated iteratively. The assignment and update cycles continue until the centroids stabilize and cease to shift significantly, signifying convergence. The objective is to reduce the total sum of squared distances between each point and its respective centroid, expressed as the following equation:

$$\sum_{i=1}^{K}\sum_{z\in S_i} \|z-\mu_i\|^2 \tag{16}$$

where S_i represents the members of cluster i and μ_i is the cluster's centroid.

2) Agglomerative Clustering: Agglomerative clustering is a hierarchical technique that builds clusters using a bottom-up approach. Initially, each data point is considered as an individual cluster. The algorithm then repeatedly merges the closest pairs of clusters based on a chosen distance metric until all points are merged into a single cluster or until a specified number of clusters is reached. This process is generally visualized using a dendrogram, which illustrates the sequence of merges and the distance at which each merge occurred.

3) Gaussian Mixture Model (GMM): GMM is based on probability, assuming that all data points arise from a combination of several Gaussian distributions, each characterized by unknown parameters. The process starts with an initial guess of the parameters of each Gaussian component, which includes means, covariances, and mixture weights. The expectation-maximization (EM) algorithm is then used in two main steps: the expectation step (E-step), where the model estimates the probability that each data point belongs to a particular Gaussian component, and the maximization step (M-step), where the model changes the parameters to maximize the likelihood of the dataset given estimated probabilities. This iterative process continues until the parameters converge, meaning changes between iterations fall below a threshold, indicating the model has effectively learned the structure of the data.

DEC introduces a soft assignment approach using the Student's t-distribution as a kernel to estimate the likeness between embedded points and cluster centroids. This step improves cluster purity and helps in refining the cluster centers. The soft assignment, q_{ij}, which defines the probability of assigning sample i to cluster j, is computed as the following equation:

$$q_{ij} = \frac{\left(1 + \left\|z_i - \mu_j\right\|^2 / \alpha\right)^{-\frac{\alpha+1}{2}}}{\sum_{j'} \left(1 + \left\|z_i - \mu_{j'}\right\|^2 / \alpha\right)^{-\frac{\alpha+1}{2}}} \tag{17}$$

where μ_j are the cluster centers, and α is the degrees of freedom of the Student's t-distribution, typically set to 1.

To emphasize data points assigned with high confidence, a target distribution p_{ij} is computed from q_{ij} by normalizing the squares of q_{ij} across each cluster as the following equation:

$$p_{ij} = \frac{q_{ij}^2 / \sum_i q_{ij}}{\sum_{j'} \left(q_{ij'}^2 / \sum_i q_{ij'}\right)}. \tag{18}$$

The parameters of the encoder and the cluster centers are jointly refined to minimize the Kullback–Leibler divergence between the soft assignments q_{ij} and the target distribution p_{ij}, guiding the learning toward a desirable clustering solution as the following equation:

$$L = KL\,(P\|Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}}. \tag{19}$$

4) Performance of DEC Model: The Silhouette Score and the Calinski–Harabasz Index are used to evaluate the quality of clustering results and find the optimal number of clusters.

1) The Silhouette Score measures how similar a data point is to its cluster compared to others and focuses on individual data points and their relationship to neighboring clusters. The score ranges from –1 to 1, where close to 1 implies the well-clustered data point, close to 0 indicates the data point is close to a decision boundary between two clusters, and close to –1 shows the data point is likely in the wrong cluster [84].
2) The Calinski–Harabasz Index measures the ratio of between-cluster dispersion to within-cluster dispersion and focuses on the overall structure of the clusters, looking at their separation and compactness. This index does not have a fixed range, but higher values indicate dense and well-separated clustering [85].

III. Results

A. Holistic Analysis Results

The following grid maps with shapes represent the SOM for the telematics dataset. Each map corresponds to a model with cells, and each cell represents a neuron, and the shapes within each neuron depict different features as concurrent events clustered by the SOM. In each model, a default structure is based on the constant variables of "Speed," "Number of Hard Acceleration," and "Number of Hard Braking" placed in every cell as concurrent events generated by the SOM algorithm, marked with red circles, blue triangles, and green diamonds. Simultaneously, target variables indicated by the square shape with different colors for each model are placed on the maps. All the variables are scaled in ascending order, with distinguishable color bars, where lighter hues correspond to "low" values, and darker hues correspond to "high" values. Similarly, smaller shapes indicate "low" values, and larger shapes indicate "high" values. The arrangement and distribution of these shapes reflect the topology of the input space as captured by the SOM, revealing patterns, relationships, and clusters within the driving data. The optimal model has a tolerable overall number of "high" events for each feature. Aggressive driving can be recognized by the excessive numbers of "high" events for a feature individually in each model and for adjacent features aggregated in each cell.

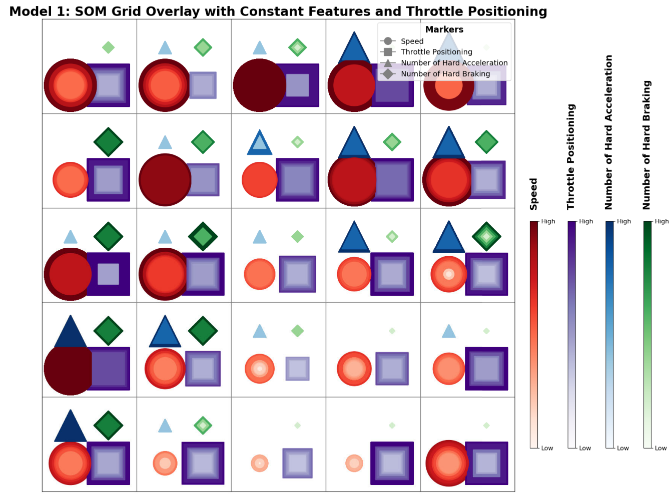

Fig. 7. Model 1 - SOM for throttle positioning overlaid with speed, hard acceleration, and hard braking.

1) Model 1- SOM for Throttle Positioning: Model 1 represents the target variable of "Throttle Positioning" using purple square shapes (see Fig. 7). In this model, cells with large circles and squares suggest scenarios where high speed coincides with significant throttle positioning, which can indicate aggressive driving or situations requiring sustained high speed, such as highway driving. Large and dark triangles, diamonds, and squares in the same cells suggest frequent hard acceleration and braking. This pattern might be observed in urban driving with stop-and-go traffic prone to congestion or in situations requiring reactive driving maneuvers, putting more stress on the vehicle's engine. Regions of the grid with light and smaller shapes might represent more conservative driving conditions with lower speeds and less aggressive throttle use.

2) Model 2- SOM for Engine Load: Model 2 displays the state of the target variable "Engine Load" using black square shapes (see Fig. 8). In this model, high engine load (large, dark squares) often appears alongside high speed (large, dark

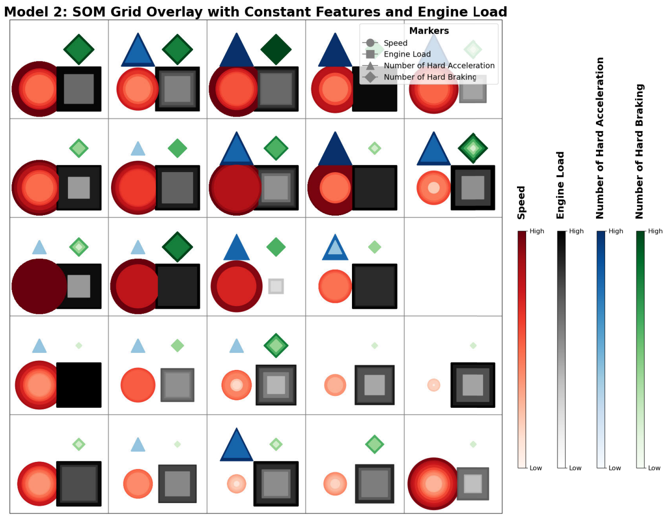

Fig. 8. Model 2 - SOM for engine load overlaid with speed, hard acceleration, and hard braking.

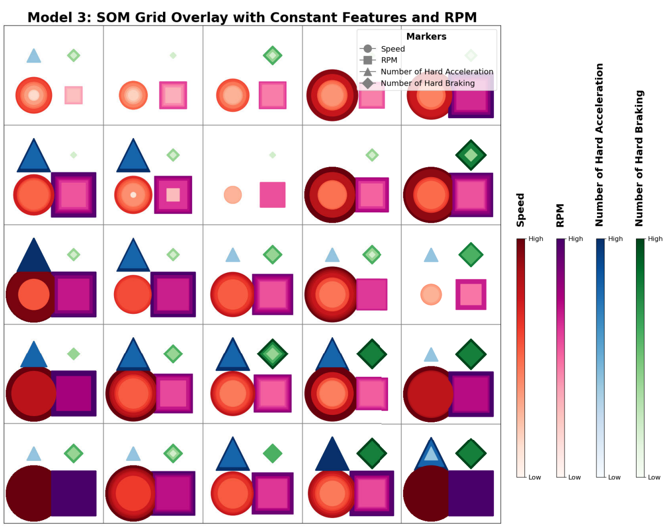

Fig. 9. Model 3 - SOM for RPM overlaid with speed, hard acceleration, and hard braking.

circles), presenting scenarios possibly involving highway driving or situations requiring high power and speed, such as uphill driving. The combination of a dark and large square with a dark and large triangle and circle in a cell could be due to scenarios requiring rapid increases in speed, such as merging onto fast-moving traffic lanes or overtaking. On the other hand, the same situation with a dark and large diamond might occur in driving scenarios where both high power output and sudden stops are common, conceivably in urban environments with fluctuating traffic conditions or in aggressive driving situations.

3) Model 3- SOM for RPM: Model 3 provides the SOM for the target variable of "RPM" as revolutions per minute (see Fig. 9). In this model, cells with large and dark circles and squares indicate high speed accompanied by high RPM, where vehicles might be driven aggressively or in conditions necessitating higher power, such as highway driving or uphill routes. When large and dark triangles (Hard Accelerations) appear with dark and large circles and squares, it suggests aggressive acceleration practices, possibly for overtaking or

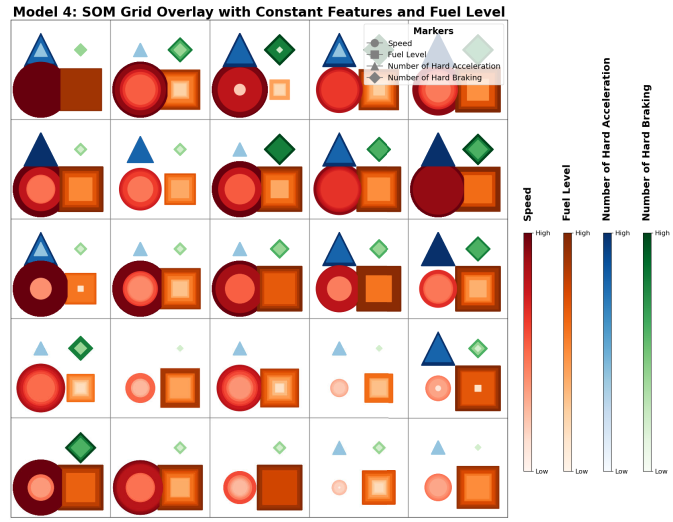

Fig. 10. Model 4 - SOM for fuel level overlaid with speed, hard acceleration, and hard braking.

quick merging. Large and dark diamonds (Hard Braking) are concerning when paired with high speed and high RPM, where drivers often need to stop suddenly, which could be indicative of high-speed urban driving or potentially hazardous driving habits. Conversely, cells with lighter and smaller shapes across all metrics could suggest more conservative driving behaviors, usually found in controlled environments like residential areas with strict speed limits.

4) Model 4- SOM for Fuel Level: Model 4 shows the SOM for the target variable of "Fuel Level" (see Fig. 10). In this model, observations of dark and large circles alongside dark and large squares (fuel level) can provide scenarios where higher speeds are maintained with a sufficient fuel supply, which might indicate long-distance travel or efficient highway driving. Alternatively, high speed with a lower fuel level might indicate less efficient fuel usage or scenarios where drivers are pushing the limits of their fuel range. The presence of dark and large triangles with varying sizes of squares can provide insights into how aggressive driving impacts fuel consumption. Frequent hard accelerations with high fuel levels suggest less concern for fuel economy, whereas similar behavior with low fuel levels could indicate inefficient driving habits that lead to frequent refueling. Dark and large diamonds appearing with different sizes of squares can highlight safety concerns. Frequent hard braking, especially with low fuel levels, might occur in urban environments where frequent stops drain fuel more quickly, indicating inefficient driving patterns.

5) Model 5- SOM for Ambient Air Temperature: Model 5 depicts the state of the target variable of "Ambient Air Temperature" (see Fig. 11). In this model, dark and large circles alongside dark and large squares (Ambient Air Temperature) might indicate that higher speeds are more common at higher temperatures. The presence of dark and large triangles with varying colors and sizes of squares indicates that hard accelerations can be affected by temperature changes. Dark and large diamonds in conjunction with various squares might reflect that braking performance or driver reaction times can

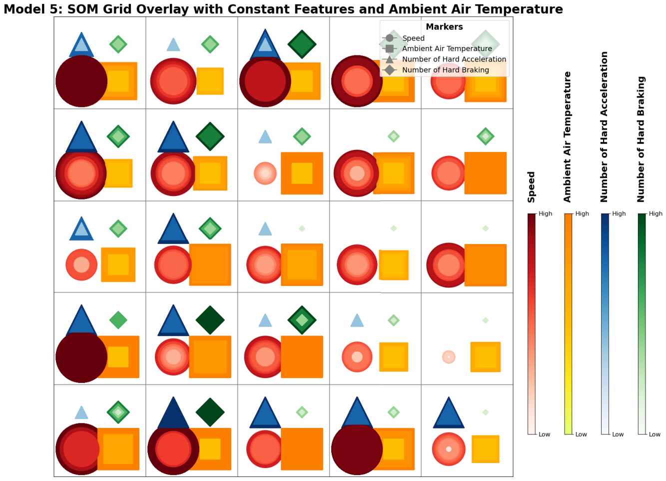

Fig. 11. Model 5 - SOM for ambient air temperature overlaid with speed, hard acceleration, and hard braking.

be affected under temperature swings, possibly due to environmental factors.

B. Component-Level Analysis Results

1) Driving Behavior Classification Based on SOM Results: Table III presents the performance metrics of quantization and TEs derived from SOMs to determine the most optimized grids of 5×5 consisting of 25 neurons as initial clusters across five distinct models to visualize the preliminary characteristics of the scaled dataset. Subsequently, we conducted the U-matrix evaluation of each model on the 5×5 grids to determine the optimal number of clusters, shown in Table IV. U-matrix visualization in Fig. 12 represents the distance between neurons in the trained SOM for each model to understand the clustering and topology of the data in a high-dimensional space mapped onto a 2-D grid. The gradient from lighter colors to darker colors indicates the distance between neurons, where lighter areas with lower values show clusters of similar items, and darker areas with higher values indicate boundaries between different clusters. The map's topology, indicated by how the clusters are laid out and separated by higher values, presents complex relationships and separations within the dataset. The contours or the values on the proposed U-matrices indicate the magnitude of the distance between neurons at those points to quantitatively assess the differences between neighboring neurons. For example, a leap from a lower value to a higher one indicates a significant difference in the feature space, suggesting a boundary between clusters. Each cluster corresponds to a unique driving behavior defined by specific features tailored to each model. Additionally, Table IV and Fig. 13 report the percentage contribution of each cluster, providing insights into the relative intensity of significant driving values within the data about older drivers.

1) *Model 1* outlines characteristics for two clusters identified in a dataset, using "Throttle Positioning" as a focal metric alongside other vehicular dynamics. Cluster 1 of this model represents scenarios where vehicles generally operate at higher speeds. A relatively high throttle positioning indicates that vehicles in this cluster often drive more aggressively, likely requiring higher power output. A higher frequency of hard accelerations supports the idea of an aggressive driving style, correlating with high throttle use. Similarly, frequent hard braking events suggest a dynamic driving environment, possibly involving rapid speed changes and reactive maneuvers. This cluster accounts for approximately 32.21% of the data, indicating a significant proportion of driving conditions within the observed dataset. Cluster 2 of this model is the opposite of Cluster 1 and reflects urban or residential driving where speeds are descending and driving actions are more deliberate and cautious. This cluster represents a larger data share at approximately 43.68%, suggesting moderate driving behaviors are more common among the observed drivers.
2) *Model 2* reflects two clusters identified in the dataset, with "Engine Load" as a primary attribute alongside other features. Cluster 1 for this model has a moderate engine load and a conservative driving style with low to moderate values for constant features. This cluster constitutes approximately 64.63% of the dataset, indicating that this driving behavior is the most common among the observed vehicles. Cluster 2 characterizes moderate engine load and high numbers of hard accelerations,

TABLE III
SOM MODEL PERFORMANCE

SOM Models	Best Performance Metrics with 5x5 grid (25 Neurons)		Optimized U-matrix Threshold	Data Included (%)	Optimized Number of Clusters	Processing Time (sec)	Memory Consumption (MB)
	Quantization Error	Topographic Error					
Model 1	0.261	0.274	0.7	75.890	2	69.58	983.80
Model 2	0.263	0.286	0.6	78.602	2	69.48	983.42
Model 3	0.217	0.256	0.65	66.797	2	69.42	983.03
Model 4	0.259	0.224	0.6	62.416	2	68.81	983.82
Model 5	0.254	0.249	0.6	59.045	3	68.99	983.58

TABLE IV
EXTRACTED DRIVING BEHAVIOR BASED ON SOM CLUSTERING RESULTS

Models	Target Variable	Clusters	Percentage (%)
Model 1	Throttle Positioning	- Cluster 1: a cautious or conservative driving pattern, likely in urban settings with lower speeds and minimal sudden movements - Cluster 2: aggressive or spirited driving behaviors with frequent and sharp changes in speed and throttle usage.	32.21 43.68
Model 2	Engine Load	- Cluster 1: conservative driving patterns with moderate engine load and lower speeds, reflecting cautious and efficient driving likely in urban driving - Cluster 2: a driving style that involves higher speeds with moderate engine load adjustments and frequent accelerations, representative of highway or rural road driving	64.63 13.96
Model 3	RPM	- Cluster 1: a low-speed, low RPM driving with occasional hard accelerations, suitable for parking lots, dense urban traffic, or smooth driving with occasional stops - Cluster 2: aggressive driving behaviors with high speed and high RPM behaviors likely on highways or in less regulated environments	51.99 14.80
Model 4	Fuel Level	- Cluster 1: a conservative, slower-speed driving with minimal sudden driving actions, indicative of urban driving soon after refueling with frequent stops and starts due to traffic congestion - Cluster 2: a scenario with low fuel levels and intense driving behavior with higher constant variables	57.82 4.58
Model 5	Ambient Air Temperature	- Cluster 1: faster drivings in lower temperate conditions - Cluster 2: lower speeds with moderate temperatures - Cluster 3: Very high hard acceleration which points to aggressive driving behavior with higher temperatures	19.16 37.71 2.16

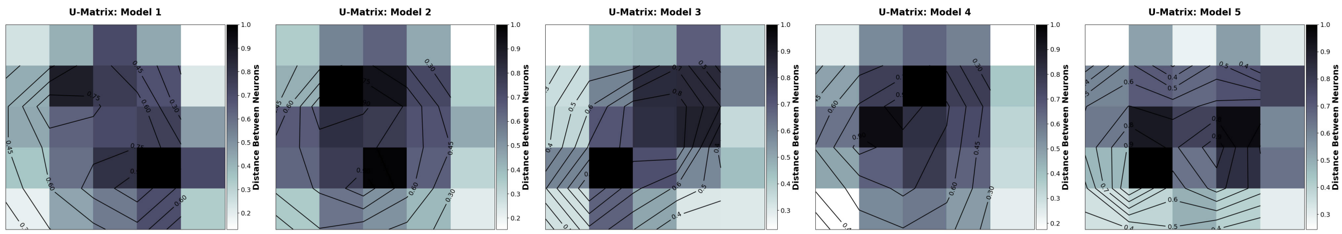

Fig. 12. U-matrix visualization for all the models.

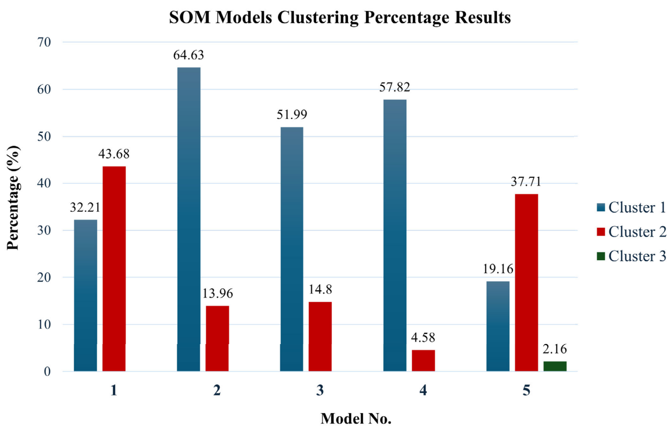

Fig. 13. SOM models clustering results.

tending to an aggressive driving style and representing only about 14% of the data as a less common driving style. Model 2 reflects two clusters identified in the dataset, with "Engine Load" as a primary attribute alongside other features. Cluster 1 for this model has a moderate engine load and a conservative driving style with low to moderate values for constant features. This cluster constitutes approximately 64.63% of the dataset, indicating that this driving behavior is the most common among the observed vehicles. Cluster 2 characterizes moderate engine load and high numbers of hard accelerations, tending to an aggressive driving style and representing only about 14% of the data as a less common driving style.

3) *Model 3* presents two clusters with "RPM" as a significant variable, reflecting different driving styles and engine performance characteristics. Cluster 1 has a low speed, suggesting driving conditions that involve slower speeds, such as urban traffic or congested areas. Corresponding to the lower speeds, the RPM is also moderate, indicating that the engine is not heavily utilized and points to driving scenarios requiring frequent stops and starts. Very low occurrences of hard accelerations in this cluster reflect a gentle driving style on the engine, minimizing rapid speed changes. The relatively low frequency of hard braking is consistent with cautious driving or driving in traffic where speeds are reduced. This dominant driving pattern covers approximately 52% of the dataset, suggesting that more than half of the observed drivers prefer a conservative driving style that avoids high speeds and excessive engine load. Meanwhile, Cluster 2 indicates higher values for all the features in this model and represents only about 15% of the data, which implies that fewer drivers engage in or have conditions that allow for high-speed and high-RPM driving.

4) *Model 4* shows two clusters with "Fuel Level" as a key variable, indicating different driving behaviors related to fuel level variations. For Cluster 1, lower values for this cluster with the moderate fuel level indicate routine use, where the vehicle is used regularly but not necessarily for extended trips that would require a full tank. Making up nearly 58% of the dataset, the driving pattern of this cluster is prevalent, suggesting that most driving occurs under conditions that do not demand frequent refueling, supporting moderate usage. Cluster 2 for this model represents higher values for constant variables and lower values for fuel level, accounting for a small portion of the data, about 4.6%, indicating that high-speed and intense driving behavior is not common among the observed vehicles.

5) *Model 5* divides driving behavior into three clusters based on various driving metrics, including the "Ambient Air Temperature," which can influence driving conditions and vehicle performance. Cluster 1 represents moderate to high speeds, suggesting driving that could be on highways or less congested areas. A lower temperature in this cluster might indicate that these data are collected in cooler climates or during cooler times of the year. A moderate frequency of hard accelerations and hard braking might imply an aggressive or dynamic driving style or situations that occasionally require quick acceleration or stops. This pattern is related to 20% of the dataset but is not predominant and only happens occasionally. In Cluster 2, reduced speed indicates urban driving or congested traffic conditions where high speeds are not possible. In this cluster, moderate temperatures could imply driving during milder weather conditions, typically comfortable and safe. At the same time, a low frequency of hard acceleration and hard braking shows a cautious driving style, likely in urban or congested driving conditions with frequent stops and starts. As the largest cluster, this reflects the most common driving conditions in the dataset, showing that most driving occurs under moderate temperatures and urban conditions. For Cluster 3, moderate speed reflects a mix of urban and open-road driving conditions. High temperatures could indicate that this data is gathered in hot climates or during summer months, which may affect the

TABLE V
DEC MODEL PERFORMANCE

DEC Models	Clustering Methods	Performance Metrices		Optimized Number of Clusters	Processing Time (sec)	Memory Consumption (MB)
		Silhouette Score	Calinski-Harabasz Index			
Model 1	**K-means ✓**	0.35	**11579.64**	2	220.23	28.19
	GMM	0.31	8444.60		82.50	25.16
	Agglomerative ✓	**0.37**	11257.96		53.88	23.58
Model 2	**K-means ✓**	**0.37**	**13196.76**	2	229.06	27.40
	GMM	0.35	12736.85		86.96	24.69
	Agglomerative	0.34	12534.49		51.72	21.77
Model 3	K-means	0.32	11302.89	2	224.13	27.29
	GMM	0.22	6931.35		83.73	24.35
	Agglomerative ✓	**0.40**	**13349.82**		51.02	22.34
Model 4	**K-means ✓**	**0.35**	**12012.89**	2	222.17	27.47
	GMM	0.33	11447.49		80.25	24.39
	Agglomerative	0.32	11131.44		50.59	23.38
Model 5	K-means	0.34	11100.76	2	223.23	28.22
	GMM	0.32	10566.65		87.86	24.53
	Agglomerative ✓	**0.38**	**11379.94**		51.21	24.92

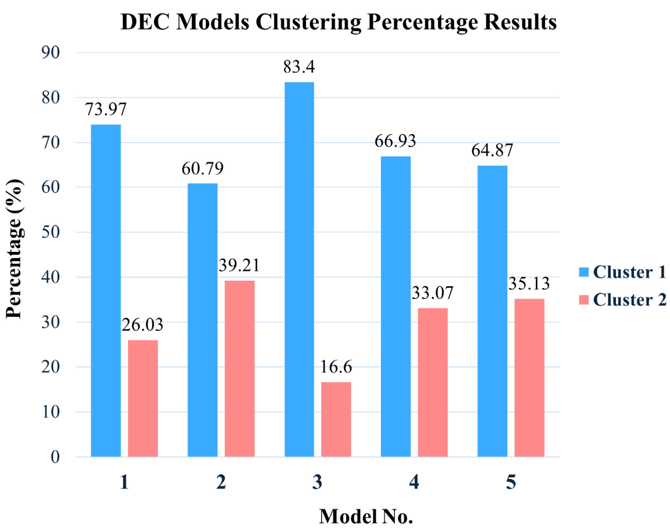

Fig. 14. DEC models clustering results.

comfort and behavior of drivers. In this cluster, very high hard acceleration points to aggressive driving behavior, potentially due to less traffic or the need to use air conditioning more aggressively in hot weather, affecting vehicle performance. Moderate hard braking, slightly higher than in Cluster 2, could indicate variable driving conditions where sudden stops are occasionally required. Cluster 3, the smallest cluster with 2.16%, indicates that hot conditions with aggressive driving behaviors are relatively rare in the dataset.

2) Driving Behavior Classification Based on DEC Results: Table V and Fig. 14 present the performance evaluation of various DEC models using different clustering methods. The assessment is conducted using the Silhouette Score and the Calinski–Harabasz Index to identify the optimal clustering method and the corresponding optimized number of clusters for each model. The clustering method that yielded the best results, as indicated by performance metrics, is marked with a check mark. The optimal number of clusters is determined for each model by computing these metrics across multiple cluster configurations. Table VI and Fig. 15 shows the results from the DEC analysis. The following is the breakdown and interpretation of the obtained clusters from this analysis.

TABLE VI
EXTRACTED DRIVING BEHAVIOR BASED ON DEC RESULTS

Model	Target Variable	Clusters	Percentage (%)
Model 1	Throttle Positioning	- Cluster 1: a typical urban or controlled-speed driving with moderate throttle usage and fewer instances of hard stops and starts	73.97
		- Cluster 2: a highway driving or situations requiring more frequent speed adjustments and more aggressive driving behavior	26.03
Model 2	Engine Load	- Cluster 1: typical of urban environments with lower power needs	60.79
		- Cluster 2: more demanding driving conditions, influencing fuel consumption, emissions, and wear and tear on the vehicle	39.21
Model 3	RPM	- Cluster 1: a driving condition with more wear and tear or higher fuel consumption due to the frequent high RPMs and speed variations	83.40
		- Cluster 2: a driving different types of mechanical stresses due to prolonged operation at low speeds and RPMs, impacting aspects like engine efficiency and emissions	16.60
Model 4	Fuel Level	- Cluster 1: a driving condition utilized in a way that prioritizes readiness with higher fuel levels, suitable for urban settings with unpredictable needs for mobility	66.93
		- Cluster 2: a driving condition operating often at lower fuel levels and higher speeds, might reflect usage patterns such as regular long-distance commuting or commercial vehicle operations where fuel optimization and time efficiency are critical	33.07
Model 5	Ambient Air Temperature	- Cluster 1: warmer conditions, might need considerations for cooling and efficiency at lower speeds	64.87
		- Cluster 2: a driving condition that vehicles may require focus on performance and safety at higher speeds and during cooler ambient temperatures.	35.13

The DEC results for *Model 1* with the target variable of "Throttle Positioning" provide insightful distinctions between the two primary clusters. For Cluster 1, the average speed is lower at around 19.8 km/h with a moderate standard deviation of 10.3, suggesting variability within the cluster. The speeds range from a minimum of 0 km/h (indicating instances of idling or stopping) to a maximum of about 47.2 km/h. The average throttle positioning is 17.2%, ranging from 7.28% to 28.94%, with a relatively low standard deviation of 5.4, indicating a more consistent but moderate level of throttle engagement across the cluster. This cluster shows fewer hard accelerations and moderate hard braking, suggesting more cautious or conservative driving behavior in this cluster. This cluster represents a more significant segment of the data, comprising about 74%.

Cluster 2 in this model has a higher average speed at approximately 47.3 km/h with a similar range of variability as Cluster 1. The speeds extend to 65.76 km/h, indicating that this cluster contains the most high-speed events. A slightly higher average throttle usage of 21.3%, with a maximum equal to Cluster 1, indicates generally more aggressive driving regarding speed and throttle usage. In this cluster, the higher frequency of hard accelerations and hard braking suggests more dynamic or aggressive driving patterns, including rapid speeding up and slowing down. This cluster has the smallest proportion of the dataset, making up about 26% of the instances, but potentially indicating more aggressive or varied driving behaviors.

The DEC results for *Model 2* with the target variable of "Engine Load" offer a detailed view into two distinct clusters that highlight differences in driving patterns based on engine load and constant variables. Cluster 1 in this model exhibits a lower average speed (about 17.57 km/h) with a standard deviation of 9.8, indicating moderate speed variability. The speed ranges from 0 to 48.4 km/h, which might suggest urban driving or areas with frequent stops. The average engine load is 29.08%, with a standard deviation of about 8.45%, showing relatively stable engine performance. The engine load range (8.64%–57.40%) indicates that most vehicles in this cluster operate under lighter load conditions, typical of city driving or lower speed conditions. This cluster reflects a lower frequency of hard acceleration and braking, correlating with more conservative driving behaviors or environments that require less aggressive driving. This cluster contains the majority of data points (about 60.79%), indicating that lower speeds and engine loads characterize most observed driving conditions.

Cluster 2 in this model, with higher average speeds (about 41.55 km/h) and broader variation in speeds (up to 65.76 km/h), suggests highway or rural driving where higher speeds are typical. A higher average engine load at 38.87%, with a narrower data spread, implies more consistent higher load operations. This cluster likely involves vehicles operating under conditions that require more power, such as inclines, heavier vehicles, or towing scenarios. Higher values for acceleration and braking in this cluster suggest more dynamic driving, likely due to faster speeds or conditions requiring frequent speed and power changes. This cluster, making up about 39.21%, represents conditions of higher demand for the vehicle.

The DEC results for *Model 3* with the target variable of "RPM" delineate two distinct clusters with the following description. Cluster 1 in this model exhibits a higher average speed of approximately 29.81 km/h with a considerable standard deviation of 14.78, suggesting varied driving conditions, including highway or urban driving with moderate to high speeds. The average RPM in this cluster is significantly higher, indicating vehicles operating at higher engine speeds, which could correspond to driving at higher gears or aggressive driving behavior. A noticeable amount of hard acceleration and braking could be linked to driving in traffic-heavy areas or environments requiring frequent stopping and starting. Cluster 1 contains the most data points (about 83.4%), indicating that most observed driving behaviors are associated with higher RPMs and, correspondingly, higher speeds.

In Cluster 2 of this model, the significantly lower average speed of about 12.70 km/h, with a high standard deviation of 14.78 relative to the mean, indicates highly variable but generally low-speed driving conditions such as congested city driving or residential areas. This cluster shows a much lower average RPM at approximately 651 RPM. The range from 508 to 841 RPM suggests mostly idle to low-speed driving, likely involving lower gear usage and less engine load. In addition, minimal hard acceleration and braking in this cluster are typical of cautious driving or congested areas where high speeds are not feasible. Cluster 2, with about 16.6% contribution, represents conditions typically associated with lower speeds and RPMs.

The DEC results of *Model 4* for the target variable "Fuel Level" elucidate two distinct clusters, which appear to highlight divergent patterns related to fuel levels, driving speeds, and driving behaviors. Cluster 1 of this model exhibits a lower average speed of about 22.55 km/h with a standard deviation of 12.62 km/h, suggesting generally slower or more variable speed conditions, likely urban or congested environments. This cluster has a notably higher fuel level at approximately 76.57%, with a relatively narrow spread of values (the standard deviation of 12.41%), suggesting vehicle operations with more fuel in the tank. In this cluster, moderate hard acceleration and braking indicate active driving, but not as aggressive as Cluster 2. Cluster 1 comprises about 66.93% of the data points, indicating that most observed events are associated with higher fuel levels and relatively moderate driving speeds.

Cluster 2 of this model demonstrates a higher average speed of approximately 35.93 km/h, with a higher standard deviation of 18.30 km/h, indicating more varied driving conditions that might include faster roads or highways. A lower average fuel level at about 48.07%, with a broader range of values (std. of 17.58%) in this cluster, suggests a pattern of operating vehicles at lower fuel levels, possibly indicating longer travel distances between refueling. In addition, more frequent hard acceleration and a higher hard braking point to a more aggressive driving style than Cluster 1. Cluster 2, about 33.07%, highlights behaviors that might correlate with lower fuel levels and higher speeds. For *Model 5*, DEC results for the target variable "Ambient Air Temperature" delineate two clusters that contrast driving speeds, ambient temperature experiences, and driving behaviors. Cluster 1 of this model represents slower-moving vehicles with an average speed of 17.80 km/h and a standard deviation of 9.39 km/h, indicating lower speed conditions are likely in urban or congested environments. The average ambient air temperature for this cluster is relatively higher at about 29.02 °C, suggesting that these vehicles might be operating in warmer conditions or during warmer times of the day. Driving with relatively higher hard accelerations and braking events may correlate with slower speeds and potentially more traffic-dense environments. Cluster 1 comprises about 64.87% of the data, indicating that a majority of the observations fall into the slower speed and higher temperature category.

On the other hand, Cluster 2 features higher speeds, with an average of 43.91 km/h and a lower standard deviation of 11.22 km/h, suggesting these vehicles are likely traveling on faster roads or highways. The slightly lower temperature at about 27.93 °C could be indicative of nighttime travel or cooler days, which might correlate with the increased speeds observed. More aggressive driving is apparent in this cluster, with a higher average of hard accelerations and braking incidents. The increased occurrence of these events could be linked to highway driving, where speeds and the need for sudden speed adjustments are frequent.

In addition, model performance metrics include running time (seconds) and memory consumption (MB). We evaluate running time and memory consumption across three clustering approaches (K-means, GMM, and agglomerative clustering) combined with DEC and SOM. K-means exhibited

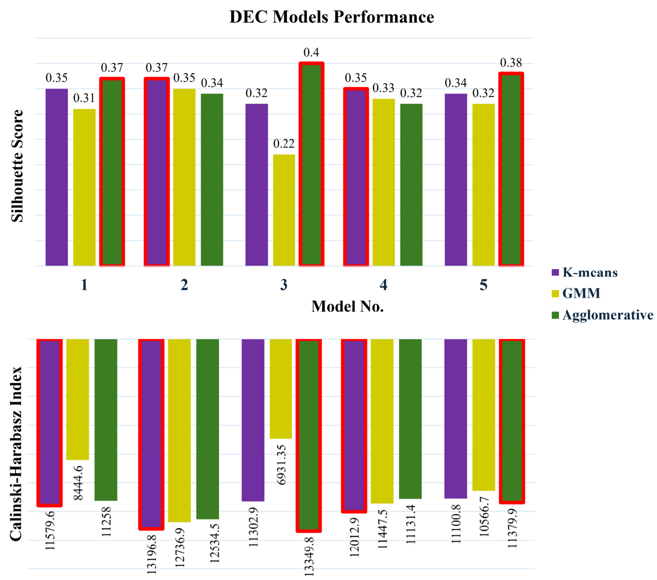

Fig. 15. DEC models performance analysis plotting.

the highest computational load (average 223 s) due to repeated recalculations of centroids, which, although accurate, pose challenges for real-time applications. GMM and agglomerative clustering required considerably less computational time and similar lower memory usage, making them more applicable for quicker deployments and real-time analysis scenarios. SOM models offered fast visualization with relatively low computational time. However, they required higher memory due to graphical processing demands, making them well-suited for preliminary real-time screening and visualization tasks.

As a result, both SOM and DEC are instrumental in analyzing older adults' complex driving behavior data, yet each method exhibits distinct strengths and weaknesses. A notable limitation of SOMs in our application is their dependency on the U-matrix to cluster data, which could not include 100% of the dataset due to optimized threshold constraints, potentially omitting subtle but important behavioral patterns. On the other hand, DEC provides a deeper understanding of the underlying structures within the driving behavior of older adults. However, DEC's implementation came with the drawback of longer processing times compared to SOM, which could be a significant hindrance in scenarios requiring rapid data analysis, such as real-time monitoring systems. Both methods thus present a trade-off between performance and depth of analysis. Although SOM provides a faster and more accessible way to handle and visualize data, DEC offers a more detailed and theoretically sound approach to understanding data clusters. Future enhancements in these methods could aim to integrate both robustness the performance of DEC and enhance the clustering inclusivity of SOM—to better support the dynamic needs of telematics data analysis.

IV. Discussion

This research focuses on assessing the driving behavior of older individuals, precisely those aged 65 and older, by applying unsupervised learning techniques. The aim is to categorize and dissect the nuances of their daily driving habits, identifying patterns and potential vulnerabilities. This detailed analysis is intended to develop strategies that improve safety for this demographic. Exploring driving patterns using unsupervised learning techniques remains a relatively untapped area of research. Identifying the most effective clustering algorithm that delivers optimal performance and determining the ideal number of clusters continues to pose significant challenges. Our study is designed to develop a framework that detects patterns in driving styles using a noninvasive, in-vehicle sensing system, leveraging the most recent advancements in machine learning technology.

We analyze telematics sensor data gathered from vehicle drivers using two distinct methods: SOM and DEC. Each method aims to determine the optimal number of clusters that best represent distinct driving patterns derived from the driving data. Both approaches identify two primary clusters or patterns within the datasets. Moreover, SOM provides the additional advantage of visually displaying overall driving patterns on maps, offering an immediate, graphical representation of the data for preliminary insights. SOM demonstrated improved performance with a structure of a 5×5 grid for each model, using the U-matrix to visualize the clustering results. DEC applied three distinct clustering techniques—K-means clustering, agglomerative clustering, and GMM—to determine the optimal clustering approach based on evaluating two performance metrics: the Silhouette Score and the Calinski-Harabasz Index. The findings identified K-means and later agglomerative clustering as the most effective methods for representing the clustering of each model.

On the other hand, results highlight an inverse relationship between clustering granularity and computational efficiency. While K-means provides detailed cluster accuracy, its higher running time and memory consumption could pose a significant bottleneck for real-time fleet monitoring systems. In contrast, models such as GMM and agglomerative clustering present more viable options for real-time monitoring applications due to their significantly lower computational load. Future implementations could try dimensionality reduction techniques and incremental clustering algorithms to look for deployment feasibility. Moreover, periodic batch processing may mitigate computational constraints while maintaining detailed analytics. These considerations should guide future telematics-based cognitive impairment detection framework integrations in large-scale and real-time operational contexts.

In our analysis of the clustering results, we find that both methodologies are highly effective: SOM excelled in providing visualizations for data interpretation, while DEC is particularly adept at mining patterns within the data. The analysis indicates that clusters associated with aggressive driving behaviors comprise a minimal portion of the dataset. Predominantly, bigger cluster percentages correspond to lower input feature values, suggesting safer driving behaviors within this demographic. However, lower values alone do not unequivocally signify safety. The behavior of other drivers plays an important role, and consistently driving below the speed limit can disrupt traffic flow and may be deemed illegal. For instance, sustained low RPM could suggest a vehicle is idling for extended periods. Moreover, driving too slowly is not without risks—it can be just as hazardous as speeding, potentially leading to traffic disruptions and accidents. In addition, external factors like weather and road conditions have a significant impact on driving patterns. Therefore, analyzing the driving conditions and external factors, identifying the selected routes, and implementing robust vehicle-to-vehicle (V2V) communications technologies is essential.

Older drivers often experience declines in sensory and cognitive functions, which can significantly affect their reaction times, decision-making, and overall driving behavior. These changes and substantial variability in how aging impacts individuals make it challenging to develop universally applicable findings and interventions. Moreover, the growing integration of advanced automotive technologies introduces additional complexity as older adults vary in their ability to adapt to and benefit from these tools due to differing levels of technological fluency. Therefore, methodologies should assess and support older drivers in effectively using such technologies. In this study, our sample predominantly includes older drivers (aged 65+) from geographically limited regions (Boca Raton, Davie, and UCF areas) recruited through convenience sampling. This approach, while practical, introduces selection bias, as participants differ from the broader older driver population in terms of demographics, socioeconomic status, or driving habits. Future research should include more diverse geographic and demographic samples to enhance the generalizability of our findings. Additionally, comparative analyses involving younger drivers or drivers from other regions could provide insight into the cross-scene applicability of our approach.

Compounding these challenges is the high dimensionality and volume of driving data collected through modern telematics systems. The complexity of this data requires advanced analytical techniques to derive actionable insights, a task for which traditional methods may fall short due to their limitations in handling simultaneous multifeature interactions and nonlinear relationships inherent in driving data. SOMs address these issues effectively by reducing the dimensionality of data while preserving the topological and metric relationships of the original dataset. This capability is particularly beneficial for analyzing telematics data, where multiple sensor readings are recorded simultaneously. SOMs project these high-dimensional data points onto a 2-D grid, making complex data structures more interpretable and allowing for the simultaneous interpretation of multiple features. This holistic overview is important for understanding the interplay between various driving behaviors, such as the relationship between speed, acceleration, and engine load, which are particularly pertinent to older adults whose driving patterns may deviate from younger drivers due to aging-related changes.

Furthermore, the graphical representation of SOMs aids in quickly identifying patterns, clusters, and anomalies without needing in-depth statistical analysis. This user-friendly visual approach provides intuitive insights accessible to researchers and practitioners who may not be experts in machine learning, highlighting areas where older drivers exhibit safe versus risky behaviors. DEC complements SOM by addressing the U-matrix method's limitations, particularly its constraints in incorporating entire datasets due to optimized threshold limitations. DEC enhances the clustering process by ensuring that all data points are effectively included and analyzed, compensating for the U-matrix method's restricted capacity to handle comprehensive dataset clustering within set threshold boundaries, which ensures that the clustering process is robust, inclusive, and reflective of the entire data spectrum.

In addition, the use of SOM in our study, although invaluable for visualizing complex driving data, presented certain limitations. Specifically, the U-matrix of SOM could not include the entire dataset in the clustering process due to the constraints imposed by the optimized threshold. This limitation suggests that some data points are overlooked in the clustering analysis, which could affect the comprehensiveness of the behavior patterns identified. Additionally, although DEC provided a robust method for assessing the coherence of driving behaviors, it is associated with longer running times than SOM. This discrepancy in running time could pose challenges in settings where rapid data processing is important. Future research could explore the development of more robust clustering algorithms that can handle complete datasets without compromising on processing speed, thereby enhancing both the accuracy and usability of the models in real-time applications.

In contrast to many studies that rely on simulated data, our research utilizes real-world driving data captured through in-vehicle telematics. This approach provides a more accurate representation of older adults' driving patterns and challenges, offering a detailed visualization of these patterns and enabling a deeper understanding of the nuanced interactions between various driving metrics. By using a combination of unsupervised learning techniques, our study identifies distinct clusters of driving behaviors among older adults and significantly advances the methodologies for studying this important demographic, ensuring that interventions are appropriately tailored to enhance safety and maintain mobility among older drivers.

This study's identification of specific driving patterns and behaviors among older drivers lays a solid foundation for future interventions and research. By distinguishing between cautious and aggressive driving tendencies, targeted training programs can be developed to address specific challenges revealed by the data. For instance, if a tendency toward high speed is observed, specialized training programs could be devised to enhance safety in complex traffic situations. Furthermore, the nuanced clustering of driving behaviors demonstrated in this study facilitates more informed policy decisions at both the municipal and state levels. With a deeper understanding of prevalent behaviors among older drivers, transportation policymakers can implement improved road designs, including enhanced signage, tailored speed limits, and intersections designed to accommodate the unique needs of older adults.

Automotive manufacturers can also benefit from the insights gained in this study to design or refine in-car systems that enhance safety for older drivers. For example, if results indicate difficulties in maintaining consistent speeds, adaptive cruise control could be emphasized in vehicles designed for older drivers. Additionally, the correlation between driving patterns and cognitive function levels offers a promising avenue for early detection of cognitive decline. Anomalies in driving behavior could serve as early indicators for health professionals to conduct check-ups or cognitive assessments, potentially allowing for earlier intervention and better management of conditions such as dementia. Addressing the challenges faced by older drivers also has the potential to mitigate the social isolation that can result from the cessation of driving. By improving and extending safe driving capabilities, older adults can maintain their independence and social connections for extended periods. Finally, the detailed analysis of driving behaviors provided by this study can assist insurance companies in developing tailored insurance packages that better meet the needs of older drivers. Healthcare providers could also use these insights during routine health visits to discuss and promote safe driving practices among older adults.

V. Conclusion

Our study advances the field of vehicular telematics by developing an unobtrusive in-vehicle sensing platform to identify driving style patterns among older drivers. This research tackles the complex challenge of integrating and analyzing data from diverse in-vehicle sensors, including GPS, IMUs, and OBD systems, using machine learning techniques. These sensors generate large volumes of data, presenting significant challenges in data processing and behavior determination. To address these challenges, our methodology employs SOMs and DEC, each chosen for its particular strengths in handling high-dimensional data. SOMs are utilized to visualize and correlate features effectively, enhancing the interpretability of complex sensor data. This visualization is crucial for understanding the nuanced driving behaviors of older adults, enabling the identification of patterns that might indicate cognitive decline or physical impairments affecting driving safety.

DEC complements this by applying advanced clustering techniques, such as K-means and GMM, to evaluate the coherence of these behaviors across the studied demographic. This approach allows us to determine the optimal number of clusters using U-matrix and various validity indices, ensuring a precise categorization of driving patterns. Such detailed clustering aids in tailoring interventions more accurately to the needs of older drivers. Furthermore, our study significantly contributes to academic research by providing a nuanced analysis of older drivers' behaviors and enhancing real-time monitoring solutions that improve road safety. By focusing on drivers aged 65 and older, and leveraging data collected over three years, we ensure that our findings are deeply relevant and specifically tailored to this demographic. The research outcomes not only facilitate a better understanding of the specific needs and capabilities of older drivers but also guide the development of more effective driver assistance technologies.

We plan to include external factors like weather and road conditions, predict and identify preferred driving routes, and establish the typical and acceptable values for driving variables specific to each route, which help ascertain whether lower observed values indicate safe or hazardous conditions for drivers. This study initially focused on older drivers aged 65 and above due to their heightened vulnerability and the disproportionate impact that accidents have on this demographic. However, the methodologies applied here are adaptable to various other demographic groups and can be tailored to different times and conditions to meet specific research objectives. Notably, the distinct ability of SOM to visually represent driving patterns on maps proves invaluable for rapid decision-making, allowing for immediate insights without the necessity for extensive data analysis using various techniques.

The current study provides initial insight into older drivers' driving behavior based on data from 51 drivers who recorded 19 881 trips with a cumulative distance of 23612941.55 km over three years. Although the data volume is substantial, expanding the participant group and extending the data collection period are planned to improve the study's reliability and breadth. The project aims to increase the sample size and diversify the data by encompassing a wider range of driving behaviors. Strict data inclusion criteria will continue to ensure data quality and reduce bias. In future endeavors, we aim to refine SOMs' visualization techniques to facilitate a simpler and more intuitive understanding of the data. This enhancement will focus on developing more user-friendly graphical representations that can easily communicate complex data relationships to both experts and nonexperts alike, improving the accessibility and applicability of our findings.

References

[1] The Administration on Aging (AoA). (2023). *2023 Profile of Older Americans*. [Online]. Available: https://acl.gov/sites/default/files/Profile%20of%20OA/ACL_ProfileOlderAmericans2023_508.pdf

[2] National Highway Traffic Safety Administration (NHTSA). *Medical Conditions*. Accessed: Jul. 5, 2024. [Online]. Available: https://www.nhtsa.gov/road-safety/older-drivers#:~:text=with%20Medical%20Conditions-,Alzheimer's%20Disease,Parkinson's%20Disease

[3] National Institute on Aging (NIA). *Safe Driving for Older Adults*. Accessed: Jul. 5, 2024. [Online]. Available: https://www.nia.nih.gov/health/safety/safe-driving-older-adults

[4] C. Brayne, C. Gill, E. S. Paykel, F. Huppert, and D. W. O'Connor, "Cognitive decline in an elderly population—A two wave study of change," *Psychol. Med.*, vol. 25, no. 4, pp. 673–683, Jul. 1995. [Online]. Available: https://api.semanticscholar.org/CorpusID:27374527

[5] L. Fraade-Blanar and J. P. Smith, "Cognitive change and driving behavior among older drivers," *Transp. Res. Record: J. Transp. Res. Board*, vol. 2672, no. 33, pp. 89–100, Dec. 2018. [Online]. Available: https://api.semanticscholar.org/CorpusID:53260539

[6] W. H. Brouwer and R. W. H. M. Ponds, "Driving competence in older persons," *Disability Rehabil.*, vol. 16, no. 3, pp. 149–161, Jan. 1994. [Online]. Available: https://api.semanticscholar.org/CorpusID:41877138

[7] J. A. A. Smither, M. Mouloua, P. Hancock, J. Duley, R. Adams, and K. Latorella, "Aging and driving Part I: Implications of perceptual and physical changes," in *Human Performance, Situation Awareness, and Automation*. Psychology Press, 2005, pp. 315–319.

[8] C. Owsley, "Driving mobility, older adults, and quality of life," *Gerontechnology*, vol. 1, no. 4, pp. 220–230, Oct. 2002. [Online]. Available: https://api.semanticscholar.org/CorpusID:108765369

[9] *Highway Safety Improvement Program Manual*. Accessed: May 22, 2024. [Online]. Available: https://safety.fhwa.dot.gov/hsip/resources/fhwasa09029/index.cfm#toc

[10] World Health Organization. *Road Traffic Injuries*. Accessed: May 22, 2024. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries

[11] E. K. Adanu and S. Jones, "Effects of human-centered factors on crash injury severities," *J. Adv. Transp.*, vol. 2017, pp. 1–11, Jan. 2017.

[12] T. K. Chan, C. S. Chin, H. Chen, and X. Zhong, "A comprehensive review of driver behavior analysis utilizing smartphones," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 10, pp. 4444–4475, Oct. 2020.

[13] M. Shahin et al., "Cluster-based association rule mining for an intersection accident dataset," in *Proc. Int. Conf. Comput., Electron. Electr. Eng. (ICE Cube)*, Oct. 2021, pp. 1–6.

[14] S. Amado, E. Arıkan, G. Kaça, M. Koyuncu, and B. N. Turkan, "How accurately do drivers evaluate their own driving behavior? An on-road observational study," *Accident Anal. Prevention*, vol. 63, pp. 65–73, Feb. 2014.

[15] N. Arbabzadeh and M. Jafari, "A data-driven approach for driving safety risk prediction using driver behavior and roadway information data," *IEEE Trans. Intell. Transp. Syst.*, vol. 19, no. 2, pp. 446–460, Feb. 2018.

[16] M. T. Jan et al., "Methods and tools for monitoring driver's behavior," in *Proc. Int. Conf. Comput. Sci. Comput. Intell. (CSCI)*, Mar. 2022, pp. 1269–1273.

[17] P. Alrassy, J. Jang, and A. W. Smyth, "OBD-data-assisted cost-based map-matching algorithm for low-sampled telematics data in urban environments," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 8, pp. 12094–12107, Aug. 2022.

[18] *GPS: The Global Positioning System*. Accessed: Jul. 5, 2024. [Online]. Available: https://www.gps.gov

[19] K. Y. Cho, C.-H. Bae, Y. Chu, and M. Suh, "Overview of telematics: A system architecture approach," *Int. J. Automot. Technol.*, vol. 7, no. 4, pp. 509–517, 2006.

[20] N. Henke, J. Bughin, M. Chui, J. Manyika, T. Saleh, and B. Wiseman, "The age of analytics: Competing in a data-driven world," *McKinsey Global Inst. Res.*, pp. 1–28, Jan. 2016.

[21] Y. Zhao, "Telematics: Safe and fun driving," *IEEE Intell. Syst.*, vol. 17, no. 1, pp. 10–14, Jan. 2002.

[22] X. Zhang, X. Wang, Y. Bao, and Z. Xiao-hui, "Safety assessment of trucks based on GPS and in-vehicle monitoring data," *Accident Anal. Prevention*, vol. 168, 2022, Art. no. 106619.

[23] S. Jafarnejad, "Machine learning-based methods for driver identification and behavior assessment: Applications for can and floating car data," Ph.D. dissertation, Dept. Comput. Sci., Fac. Sci., Technol. Commun., Interdiscipl. Centre Secur., Rel. Trust (SNT), Univ. Luxembourg, Esch-sur-Alzette, Luxembourg, 2020.

[24] M. Alsaidi, A. Saleem Altaher, M. Tanveer Jan, A. Altaher, and Z. Salekshahrezaee, "COVID-19 classification using deep learning two-stage approach," 2022, *arXiv:2211.15817*.

[25] X. Di et al., "Using naturalistic driving data to predict mild cognitive impairment and dementia: Preliminary findings from the longitudinal research on aging drivers (LongROAD) study," *Geriatrics*, vol. 6, no. 2, p. 45, Apr. 2021.

[26] M. Shuqair, J. Jimenez-Shahed, and B. Ghoraani, "Multi-shared-task self-supervised CNN-LSTM for monitoring free-body movement UPDRS-III using wearable sensors," *Bioengineering*, vol. 11, no. 7, p. 689, Jul. 2024.

[27] X. Li, W. Wang, and M. Roetting, "Estimating driver's lane-change intent considering driving style and contextual traffic," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 9, pp. 3258–3271, Sep. 2019.

[28] A. E. Abdelrahman, H. S. Hassanein, and N. Abu-Ali, "Robust data-driven framework for driver behavior profiling using supervised machine learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 4, pp. 3336–3350, Apr. 2022.

[29] S. G. Ara Ghoreishi et al., "Anomalous behavior detection in trajectory data of older drivers," in *Proc. IEEE 20th Int. Conf. Smart Communities, Improving Quality Life using AI, Robot. IoT (HONET)*, Dec. 2023, pp. 146–151.

[30] M. R. Carlos, L. C. González, J. Wahlström, G. Ramírez, F. Martínez, and G. Runger, "How smartphone accelerometers reveal aggressive driving behavior?—The key is the representation," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 8, pp. 3377–3387, Aug. 2020.

[31] X. Li, Q. Zhang, Z. Peng, A. Wang, and W. Wang, "A data-driven two-level clustering model for driving pattern analysis of electric vehicles and a case study," *J. Cleaner Prod.*, vol. 206, pp. 827–837, Jan. 2019.

[32] A. Fattahi, A. Golroo, and M. Ghatee, "Driver behavior assessment using multi-layer perceptron and random forest via smartphone sensors and OBD II," *SSRN J.*, pp. 1–33, Apr. 2023.

[33] J. Morton, T. A. Wheeler, and M. J. Kochenderfer, "Analysis of recurrent neural networks for probabilistic modeling of driver behavior," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 5, pp. 1289–1298, May 2017.

[34] P. S. Albert, "Innovative modeling of naturalistic driving data: Inference and prediction," *Statist. Med.*, vol. 38, no. 2, pp. 175–183, Jan. 2019. [Online]. Available: https://api.semanticscholar.org/CorpusID:41702261

[35] J. C. F. de Winter and R. Happee, "Modelling driver behaviour: A rationale for multivariate statistics," *Theor. Issues Ergonom. Sci.*, vol. 13, no. 5, pp. 528–545, Sep. 2012. [Online]. Available: https://api.semanticscholar.org/CorpusID:143745771

[36] N. Li, J. J. Jain, and C. Busso, "Modeling of driver behavior in real world scenarios using multiple noninvasive sensors," *IEEE Trans. Multimedia*, vol. 15, no. 5, pp. 1213–1225, Aug. 2013. [Online]. Available: https://api.semanticscholar.org/CorpusID:17822872

[37] Y. Wang and I. W.-H. Ho, "Joint deep neural network modelling and statistical analysis on characterizing driving behaviors," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2018, pp. 1–6. [Online]. Available: https://api.semanticscholar.org/CorpusID:53025582

[38] M. Siami, M. Naderpour, and J. Lu, "A mobile telematics pattern recognition framework for driving behavior extraction," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 1459–1472, Mar. 2021.

[39] E. Mantouka, E. Barmpounakis, E. Vlahogianni, and J. Golias, "Smartphone sensing for understanding driving behavior: Current practice and challenges," *Int. J. Transp. Sci. Technol.*, vol. 10, no. 3, pp. 266–282, Sep. 2021.

[40] R. L. K. Venkateswarlu and R. V. Kumari, "Novel approach for speech recognition by using self—Organized maps," in *Proc. Int. Conf. Emerg. Trends Netw. Comput. Commun. (ETNCC)*, Apr. 2011, pp. 215–222.

[41] T. Kohonen, "Self-organized formation of topologically correct feature maps," *Biol. Cybern.*, vol. 43, no. 1, pp. 59–69, 1982.

[42] A. Jalal, S. Kamal, and D. Kim, "Depth map-based human activity tracking and recognition using body joints features and self-organized map," in *Proc. 5th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2014, pp. 1–6.

[43] A. Altaher et al., "Finger knuckle print classification using pretrained vision models," in *Proc. IEEE 20th Int. Conf. Smart Communities, Improving Quality Life using AI, Robot. IoT (HONET)*, Dec. 2023, pp. 62–67.

[44] J. Qian et al., "Introducing self-organized maps (SOM) as a visualization tool for materials research and education," *Results Mater.*, vol. 4, Dec. 2019, Art. no. 100020.

[45] J. E. Geach, "Unsupervised self-organized mapping: A versatile empirical tool for object selection, classification and redshift estimation in large surveys," *Monthly Notices Roy. Astronomical Soc.*, vol. 419, no. 3, pp. 2633–2645, Jan. 2012.

[46] S. Mavroudi, S. Papadimitriou, and A. Bezerianos, "Gene expression data analysis with a dynamically extended self-organized map that exploits class information," *Bioinformatics*, vol. 18, no. 11, pp. 1446–1453, Nov. 2002.

[47] A. Mishra, B. P. Rogers, L. M. Chen, and J. C. Gore, "Functional connectivity-based parcellation of amygdala using self-organized mapping: A data driven approach," *Human Brain Mapping*, vol. 35, no. 4, pp. 1247–1260, Apr. 2014.

[48] K. Hajipour and V. Mehrdad, "Edge detection of noisy digital image using optimization of threshold and self organized map neural network," *Multimedia Tools Appl.*, vol. 80, no. 4, pp. 5067–5086, 2020.

[49] F. R. Doshi and T. Konkle, "Cortical topographic motifs emerge in a self-organized map of object space," *Sci. Adv.*, vol. 9, no. 25, p. 8187, Jun. 2023.

[50] Q.-V. Doan, H. Kusaka, T. Sato, and F. Chen, "S-SOM v1.0: A structural self-organizing map algorithm for weather typing," *Geosci. Model Develop.*, vol. 14, no. 4, pp. 2097–2111, Apr. 2021.

[51] A. Sanjurjo-De-No, B. Arenas-Ramírez, J. Mira, and F. Aparicio-Izquierdo, "Driver pattern identification in road crashes in Spain," *IEEE Access*, vol. 8, pp. 182014–182025, 2020.

[52] S. Zong, X. Wang, K. Zhang, L. Yang, and H. Shang, "Identification and difference analysis of motorcycle drivers' driving style based on natural driving data," in *Proc. Int. Conf. Intell. Transp., Big Data Smart City (ICITBS)*, Mar. 2022, pp. 301–304.

[53] S. Lakshminarayanan, "Application of self-organizing maps on time series data for identifying interpretable driving manoeuvres," *Eur. Transp. Res. Rev.*, vol. 12, no. 1, pp. 1–11, 2020.

[54] Y. Shichkina, R. Fatkieva, and M. Kopylov, "Analysis of driving style using self-organizing maps to analyze driver behavior," *Int. J. Electr. Comput. Eng.*, vol. 14, no. 2, p. 2212, Apr. 2024.

[55] J. Guo, Y. Liu, L. Zhang, and Y. Wang, "Driving behaviour style study with a hybrid deep learning framework based on GPS data," *Sustainability*, vol. 10, no. 7, p. 2351, Jul. 2018.

[56] A. Flexer, "Limitations of self-organizing maps for vector quantization and multidimensional scaling," in *Proc. Neural Inf. Process. Syst.*, vol. 9, 1996, pp. 445–451. [Online]. Available: https://api.semanticscholar.org/CorpusID:9065935

[57] A. Rauber, D. Merkl, and M. Dittenbach, "The growing hierarchical self-organizing map: Exploratory analysis of high-dimensional data," *IEEE Trans. Neural Netw.*, vol. 13, no. 6, pp. 1331–1341, Nov. 2002. [Online]. Available: https://api.semanticscholar.org/CorpusID:15510168

[58] R. Lasri, "Clustering and classification using a self-organizing MAP: The main flaw and the improvement perspectives," in *Proc. SAI Comput. Conf. (SAI)*, Jul. 2016, pp. 1315–1318. [Online]. Available: https://api.semanticscholar.org/CorpusID:18630846

[59] Y. Ren, K. Hu, X. Dai, L. Pan, S. C. H. Hoi, and Z. Xu, "Semi-supervised deep embedded clustering," *Neurocomputing*, vol. 325, pp. 121–130, Jan. 2019.

[60] J. Xie, R. B. Girshick, and A. Farhadi, "Unsupervised deep embedding for clustering analysis," in *Proc. 33rd Int. Conf. Mach. Learn. (ICML)*, Jun. 2016, pp. 478–487.

[61] M. Malik, R. Nandal, S. Dalal, V. Jalglan, and D.-N. Le, "Driving pattern profiling and classification using deep learning," *Intell. Autom. Soft Comput.*, vol. 28, no. 3, pp. 887–906, 2021.

[62] K. Yang, C. A. Haddad, G. Yannis, and C. Antoniou, "Classification and evaluation of driving behavior safety levels: A driving simulation study," *IEEE Open J. Intell. Transp. Syst.*, vol. 3, pp. 111–125, 2022.

[63] G. Li et al., "Influence of traffic congestion on driver behavior in post-congestion driving," *Accident Anal. Prevention*, vol. 141, Jun. 2020, Art. no. 105508.

[64] A. Mohammadnazar, R. Arvin, and A. J. Khattak, "Classifying travelers' driving style using basic safety messages generated by connected vehicles: Application of unsupervised machine learning," *Transp. Res. C, Emerg. Technol.*, vol. 122, Jan. 2021, Art. no. 102917.

[65] X. Zhao, Z. Wang, Z. Xu, Y. Wang, X. Li, and X. Qu, "Field experiments on longitudinal characteristics of human driver behavior following an autonomous vehicle," *Transp. Res. C, Emerg. Technol.*, vol. 114, pp. 205–224, May 2020.

[66] Y. Ma, W. Li, K. Tang, Z. Zhang, and S. Chen, "Driving style recognition and comparisons among driving tasks based on driver behavior in the online car-hailing industry," *Accident Anal. Prevention*, vol. 154, May 2021, Art. no. 106096.

[67] J. Hu, X. Zhang, and S. Maybank, "Abnormal driving detection with normalized driving behavior data: A deep learning approach," *IEEE Trans. Veh. Technol.*, vol. 69, no. 7, pp. 6943–6951, Jul. 2020.

[68] M. Shahverdy, M. Fathy, R. Berangi, and M. Sabokrou, "Driver behavior detection and classification using deep convolutional neural networks," *Expert Syst. Appl.*, vol. 149, Jul. 2020, Art. no. 113240.

[69] G. Sheng, Q. Wang, C. Pei, and Q. Gao, "Contrastive deep embedded clustering," *Neurocomputing*, vol. 514, pp. 13–20, Dec. 2022. [Online]. Available: https://api.semanticscholar.org/CorpusID:252455779

[70] U. Shaham, K. Stanton, H. Li, B. Nadler, R. Basri, and Y. Kluger, "SpectralNet: Spectral clustering using deep neural networks," 2018, *arXiv:1801.01587*.

[71] M. H. Alkinani, W. Z. Khan, and Q. Arshad, "Detecting human driver inattentive and aggressive driving behavior using deep learning: Recent advances, requirements and open challenges," *IEEE Access*, vol. 8, pp. 105008–105030, 2020.

[72] D. I. Tselentis and E. Papadimitriou, "Machine learning approaches exploring the optimal number of driver profiles based on naturalistic driving data," *Transp. Res. Interdiscipl. Perspect.*, vol. 21, Sep. 2023, Art. no. 100900.

[73] AutoPi. *The All-in-One Platform for Vehicle Data Analytics*. Accessed: Jul. 5, 2024. [Online]. Available: https://www.autopi.io/

[74] M. Farsi, M. Barbosa, and K. Ratcliff, "An overview of controller area network," *Comput. Control Eng. J.*, vol. 10, no. 3, pp. 113–120, Jun. 1999.

[75] K. H. Johansson, M. Törngren, and L. Nielsen, "Vehicle applications of controller area network," in *Handbook Networked Embedded Control Syst.*, 2005, pp. 741–765.

[76] S. Leonhardt, L. Leicht, and D. Teichmann, "Unobtrusive vital sign monitoring in automotive environments—A review," *Sensors*, vol. 18, no. 9, p. 3080, Sep. 2018.

[77] "Event data recorders-summary of finding by the NHTSA EDR working group," U.S. Dept. Transp., Nat. Highway Traffic Saf. Admin., (NHTSA), Nat. Center Statist. Anal. (NCSA), Washington, DC, USA, Tech. Rep. DOT HS 809 474, 2001.

[78] "Driver brake and accelerator controls and pedal misapplication rates in North Carolina," U.S. Dept. Transp., Nat. Highway Traffic Saf. Admin. (NHTSA), Washington, DC, USA, Tech. Rep. DOT HS 812 058, 2015.

[79] L. Boodlal and K.-H. Chiang, "Study of the impact of a telematics system on safe and fuel-efficient driving in trucks," U.S. Dept. Transp., Federal Motor Carrier Saf. Admin. Office Anal., Res., Technol., Washington, DC, USA, Tech. Rep. FMCSA-13-020, 2014.

[80] H. A. Ameen et al., "Identification of driving safety profiles in vehicle to vehicle communication system based on vehicle OBD information," *Information*, vol. 12, no. 5, p. 194, Apr. 2021.

[81] P. Li, M. Abdel-Aty, and J. Yuan, "Using bus critical driving events as surrogate safety measures for pedestrian and bicycle crashes based on GPS trajectory data," *Accident Anal. Prevention*, vol. 150, Feb. 2021, Art. no. 105924.

[82] A. Ultsch, "U-matrix: A tool to visualize clusters in high dimensional data," Dept. Comput. Sci., Univ. Marburg, Marburg, Germany, Tech. Rep. Nr. 36, 2003.

[83] J. Lötsch and A. Ultsch, "Exploiting the structures of the U-matrix," in *Proc. 10th Int. Workshop Adv. Self-Organizing Maps Learn. Vector Quantization*, Mittweida, Germany, 2014, pp. 249–257.

[84] S. Saranya and D. P. Jayanthi, "An enhanced initialization method to find an initial center for k-modes clustering," *Int. J. Recent Innov. Trends Comput. Commun.*, vol. 5, no. 1, pp. 47–53, 2017. [Online]. Available: https://api.semanticscholar.org/CorpusID:212485569

[85] T. Saelao, P. Chutimanukul, A. Suratanee, and K. Plaimas, "Analysis of antioxidant capacity variation among Thai holy basil cultivars (*Ocimum tenuiflorum* L.) using density-based clustering algorithm," *Horticulturae*, vol. 9, no. 10, p. 1094, Oct. 2023. [Online]. Available: https://www.mdpi.com/2311-7524/9/10/1094

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